

HARTFORD BRADLEY, CT, USA

WMO: 725080

Lat: 41.938N

Lon: 72.682W

Elev: 53

StdP: 100.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14740

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.6	-12.7	-23.6	0.4	-13.3	-21.0	0.6	-10.8	12.1	-2.7	11.0	-3.8	3.2	320	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	33.1	23.0	31.5	22.2	29.9	21.3	24.6	30.3	23.9	28.8	23.0	27.4	4.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	17.8	27.0	22.3	17.1	26.2	21.6	16.3	25.6	74.7	30.4	71.5	29.0	68.3	27.6	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.7	7.8	-19.4	36.1	3.0	1.6	-21.6	37.2	-23.3	38.1	-25.0	39.0	-27.2	40.2		
(5)			-20.0	26.2	2.8	1.0	-22.0	26.9	-23.7	27.5	-25.2	28.0	-27.3	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-2.8	-1.3	3.3	9.8	15.6	20.5	23.5	22.5	18.4	11.9	5.8	0.5	(6)
(7)		DBStd	10.11	5.77	5.28	5.08	4.61	4.22	3.64	2.95	3.03	3.82	4.34	4.64	4.91	(7)
(8)		HDD10.0	1454	396	317	217	57	3	0	0	0	0	29	140	295	(8)
(9)		HDD18.3	3238	654	549	467	258	109	19	1	3	46	205	374	551	(9)
(10)		CDD10.0	1716	1	1	8	52	176	314	420	387	253	87	16	2	(10)
(11)		CDD18.3	459	0	0	1	3	23	84	163	132	48	5	0	0	(11)
(12)		CDH23.3	4111	0	0	7	56	284	761	1498	1110	361	35	0	0	(12)
(13)		CDH26.7	1422	0	0	1	18	94	257	578	373	94	5	0	0	(13)
(14)	Wind	WSAvg	3.4	3.7	3.8	4.1	4.0	3.5	3.1	2.9	2.7	2.9	3.2	3.4	3.5	(14)
(15)	Precipitation	PrecAvg	1163	84	81	94	95	101	105	97	100	107	100	96	98	(15)
(16)		PrecMax	1748	244	226	174	252	305	345	284	296	285	415	217	212	(16)
(17)		PrecMin	748	10	11	7	28	22	17	25	14	21	9	10	21	(17)
(18)		PrecStd	216	48	38	38	43	54	66	53	56	68	65	44	45	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.6	15.4	22.8	29.5	32.3	34.0	35.6	34.5	32.1	27.0	20.7	16.7	(19)
(20)			MCWB	12.5	11.4	13.3	17.7	20.3	23.6	24.5	24.1	22.8	19.1	15.3	13.5	(20)
(21)		2%	DB	10.7	11.5	17.3	24.1	29.4	31.7	33.3	32.0	29.3	23.7	17.9	13.0	(21)
(22)			MCWB	8.3	7.3	10.4	14.1	19.3	22.2	23.3	23.0	21.0	17.8	14.2	10.7	(22)
(23)		5%	DB	7.4	8.5	13.8	20.8	26.5	29.6	31.7	30.3	27.2	21.6	15.8	10.4	(23)
(24)			MCWB	4.9	4.7	8.2	12.3	17.7	21.0	22.6	21.9	20.1	16.9	12.4	7.9	(24)
(25)		10%	DB	4.7	5.8	10.9	18.0	23.8	27.7	29.9	28.7	25.2	19.3	13.6	7.7	(25)
(26)			MCWB	2.5	2.9	6.7	10.9	16.0	19.7	21.6	21.0	19.0	14.8	10.0	5.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	12.7	15.3	18.6	22.4	24.9	25.8	25.5	24.0	21.6	17.8	14.2	(27)
(28)			MCD B	14.3	14.0	20.7	26.4	30.1	31.6	32.5	32.1	28.6	24.3	19.2	15.0	(28)
(29)		2%	WB	8.8	8.0	12.0	16.5	20.6	23.7	24.8	24.4	22.8	19.6	15.2	11.0	(29)
(30)			MCD B	9.9	10.6	15.5	22.2	26.5	29.4	30.8	29.4	26.8	22.3	17.1	12.3	(30)
(31)		5%	WB	5.2	5.1	9.2	14.3	19.1	22.6	24.0	23.6	21.8	17.7	12.9	8.2	(31)
(32)			MCD B	7.0	7.7	12.8	18.1	24.3	27.4	29.1	27.8	25.2	20.6	15.2	10.2	(32)
(33)		10%	WB	2.8	3.1	6.9	12.1	17.6	21.5	23.1	22.7	20.6	15.7	10.6	5.5	(33)
(34)			MCD B	4.3	5.3	10.3	16.6	22.0	25.6	27.7	26.6	23.8	18.5	12.9	7.4	(34)
(35)	Mean Daily Temperature Range			MDBR	8.4	9.1	9.9	11.5	11.8	11.3	11.2	11.4	10.9	9.4	8.0	(35)
(36)		5% DB	MCDBR	10.8	12.0	14.5	16.4	16.0	13.9	12.9	12.5	13.1	13.5	12.3	11.2	(36)
(37)			MCWBR	9.1	8.8	8.9	8.4	7.6	6.4	4.9	5.0	6.2	7.9	8.9	9.4	(37)
(38)		5% WB	MCDBR	10.5	11.1	12.9	13.4	13.7	11.7	11.0	10.5	10.3	11.4	10.8	10.6	(38)
(39)			MCWBR	9.3	8.8	8.9	8.4	7.3	6.1	4.9	4.9	5.8	8.0	9.3	9.8	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.298	0.312	0.336	0.371	0.413	0.458	0.478	0.446	0.380	0.344	0.313	0.297	(40)
(41)		taud		2.435	2.411	2.424	2.366	2.282	2.187	2.142	2.220	2.381	2.496	2.538	2.524	(41)
(42)		Ebn at Noon		852	895	912	898	864	822	801	816	852	849	832	827	(42)
(43)		Edh at Noon		77	92	103	117	131	144	150	135	107	85	70	66	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.80	2.66	3.67	4.71	5.25	5.69	5.89	5.25	4.31	2.83	1.97	1.48	(44)
(45)		RadStd		0.14	0.24	0.27	0.51	0.55	0.56	0.34	0.29	0.34	0.28	0.17	0.15	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+93	+57
(47) Regional (84 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+100	+50

Nomenclature: See separate page